Amplitude and Loudness

Intro to Audio Tech 1
Jocelyn Kavanagh
Monday, August 25

Georgia Tech

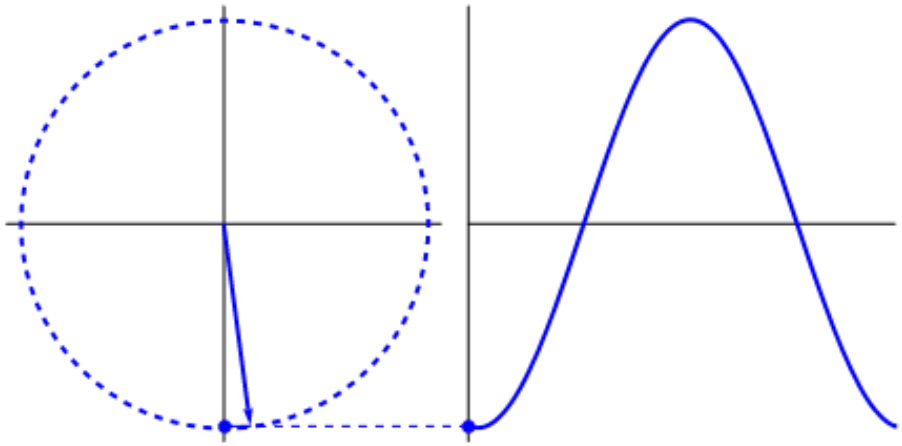
Lesson Outline

- Measures of Amplitude
- Controlling Amplitude
- Up Next…

Sinusoids

$$x[n] = a\cos(\omega n + \phi)$$

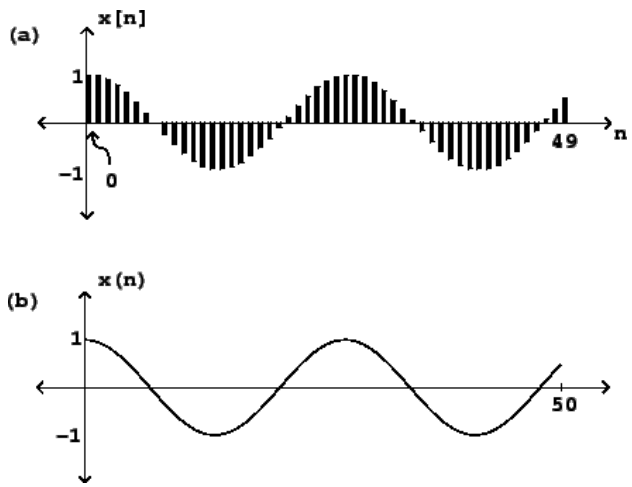
- Amplitude (decibels)
- Frequency (hertz)
- Phase

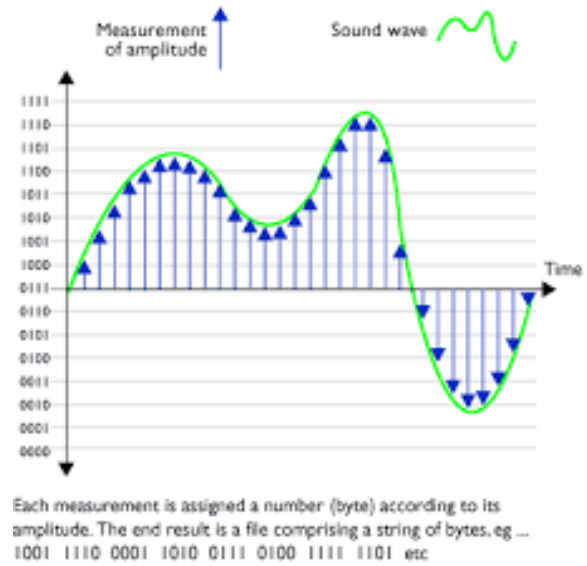


Amplitude

What is Amplitude?

- The displacement of particles in the medium from mean position
- Relative changes in pressure as the sound travels
- Can be defined as time-varying voltage or air pressure (unitless in digital world)



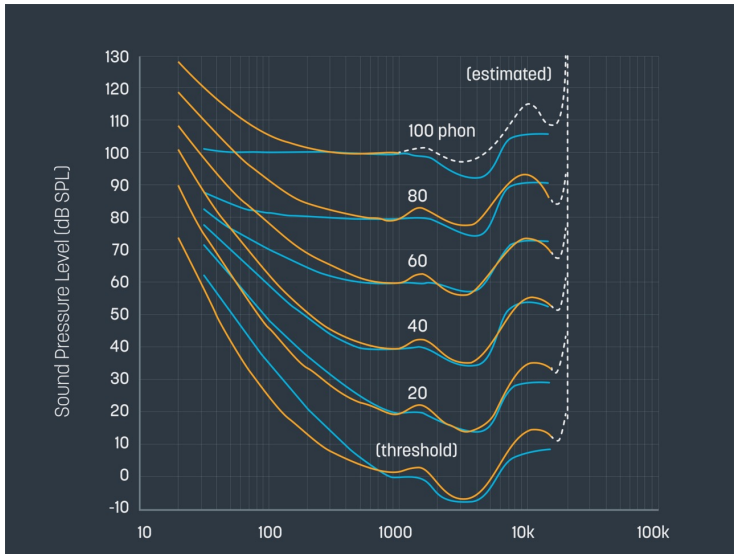


What about loudness?

- Loudness is a perceptual phenomenon
- We measure sound acoustically using:
  - Sound Pressure Level (SPL)
    - Measured in pascals
    - Depends on location and distance from the sound source
  - Sound Power Level
    - Measured in watts
    - Independent of SPL, how much energy is generated over time
  - Sound Intensity
    - Measured in watts/meters^2
    - Amount of energy transmitted over time through an area in a specified direction

Loudness – An Auditory Phenomenon

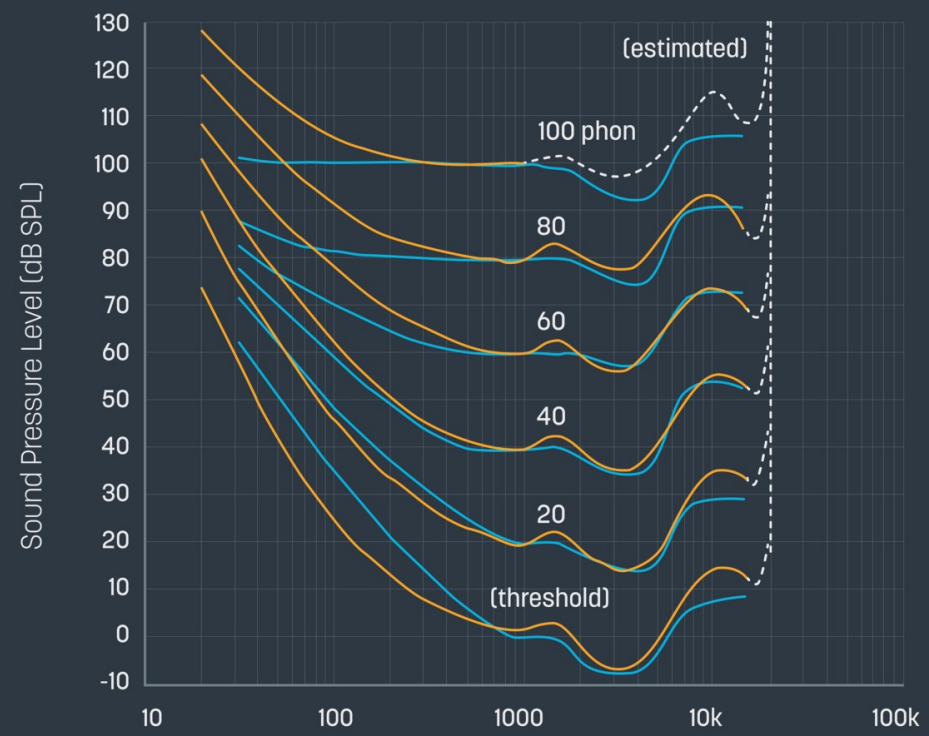
- Related to intensity but not 1-1
- Roughly logarithmic so we use decibels
- Depends on
  - Frequency
    - Fletcher Munson Curves
    - Equal Loudness Contours
  - Time
    - Constant loudness over time
    - Transients
  - “the amount of information”



Equal loudness countours (yellow) (from ISO 226:2003 revision)
Fletcher-Munson curves shown (blue) for comparison

Equal Loudness Contours

- What dbSPL does a 10k Hz tone need to be played to match a 1k Hz tone played at 20 dbSPL?



Equal loudness countours (yellow) (from ISO 226:2003 revision)
Fletcher-Munson curves shown (blue) for comparison

How are they related

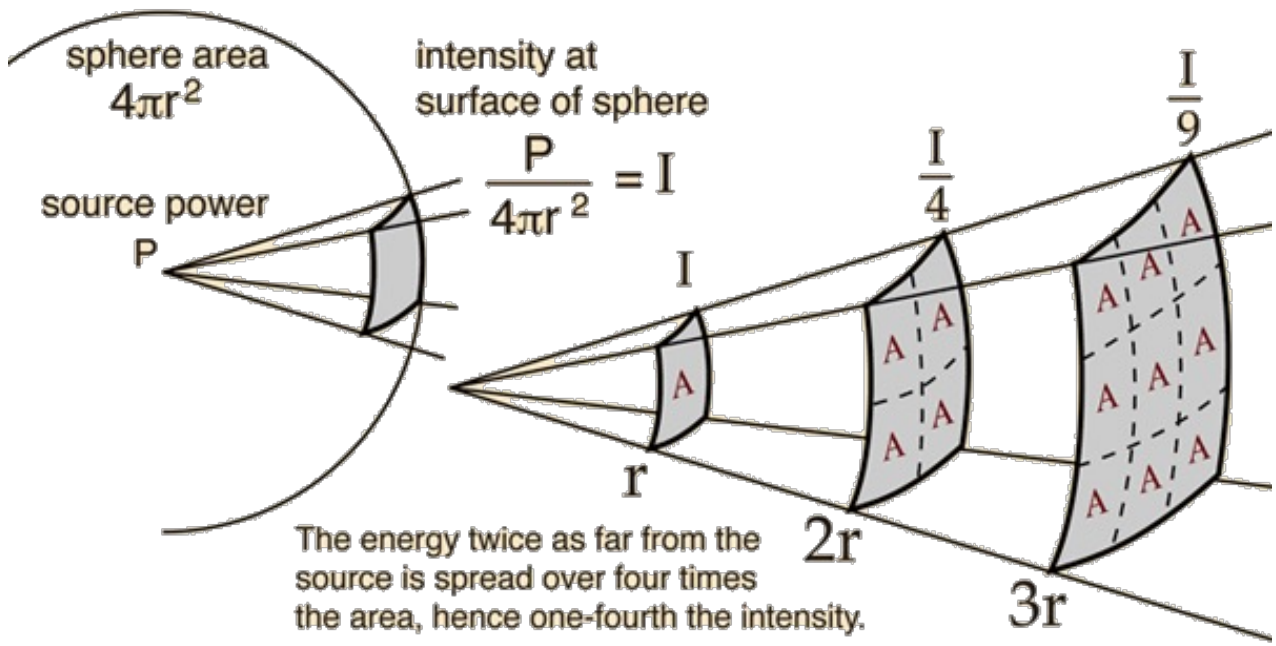
- Amplitude is the displacement of the particles
  - Can be thought of as the energy of the signal
- Power is the rate the energy is being produced (energy over time)
  - Does not change with distance
- Intensity is the power across an area
  - Combines power and distance

- Louder sounds will have larger energy, larger displacement of particles, larger intensity

Georgia Tech

Inverse Square Law

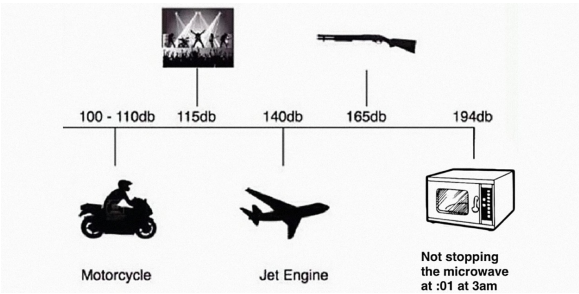
- Intensity is the power across an area



Decibels: Units of Loudness

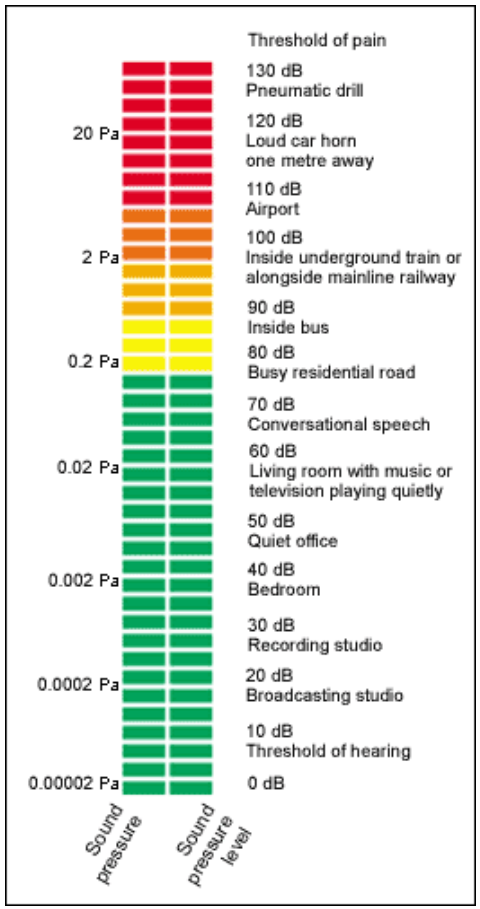
- Loudness is compared by ratio rather than direct differences
  - Why?
- Decibels
  - Logarithmic unit for ratio comparison

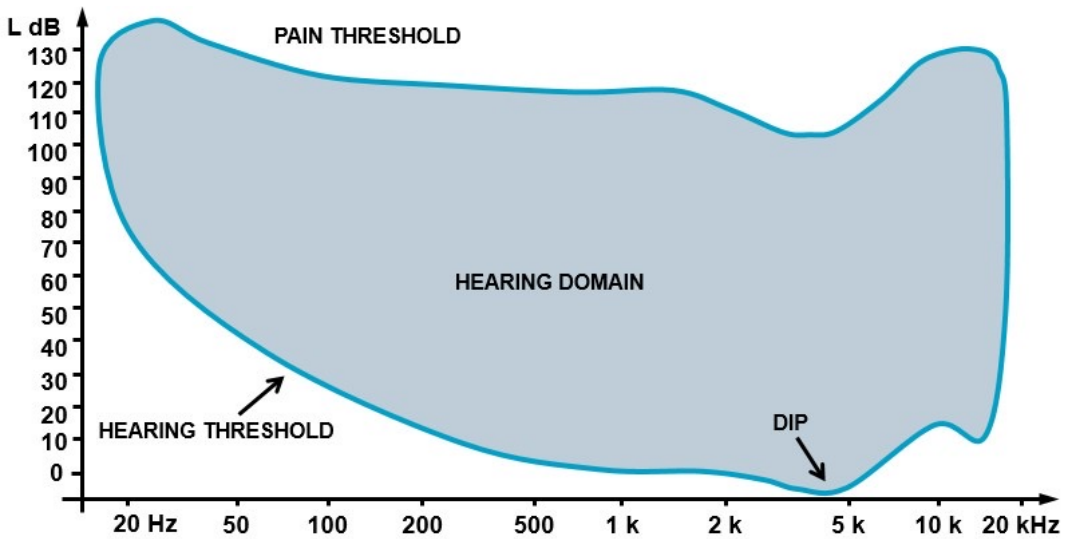
$$dB = 20 * \log(\frac{a}{a_0})$$



  - We can choose different reference amplitudes
    - $a_0 \approx 10^{-5} = 0.00001$ -- maps well to 0-1 range for digital audio
    - $a_0 \approx 10^{0} = 1$ – maps well to 0-1 range for digital audio, but negative dBFS
    - $a_0 \approx 20\ microPascals$ – 0 dB SPL is the threshold for hearing at 1 kHz

Decibel Scale





More on Decibels

- Used to measure/compare all sorts of units
  - Power, sound pressure level, sound intensity, voltage
- Log base 10 makes familiar powers of 10

dB	-12	-6	0	6	12	18	20	24
Ratio	0.25:1	0.5:1	1:1	2:1	4:1	8:1	10:1	16:1

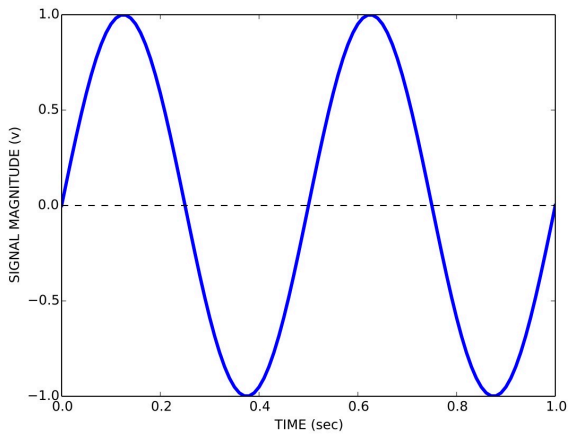
- If I say a sound is double the loudness, how many dBs did it increase?

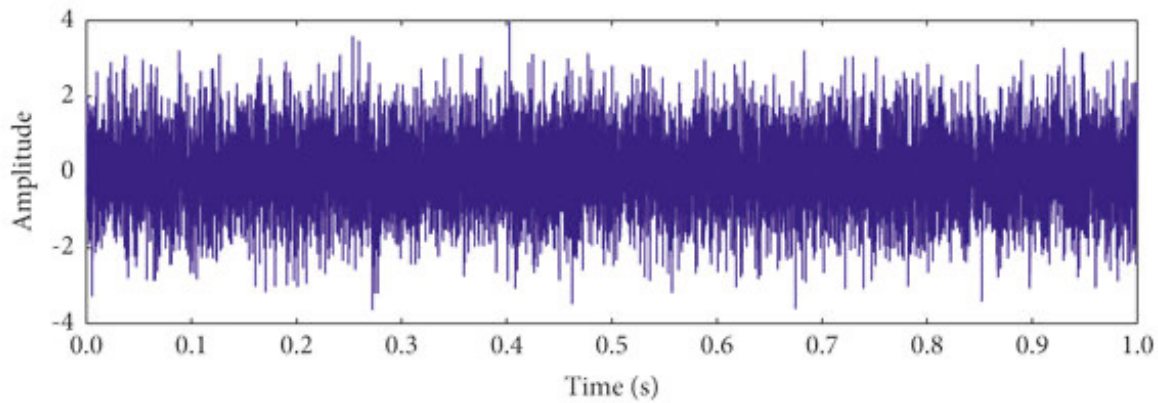
Georgia Tech

Back to Digital Amplitude

Step Back: Time Domain

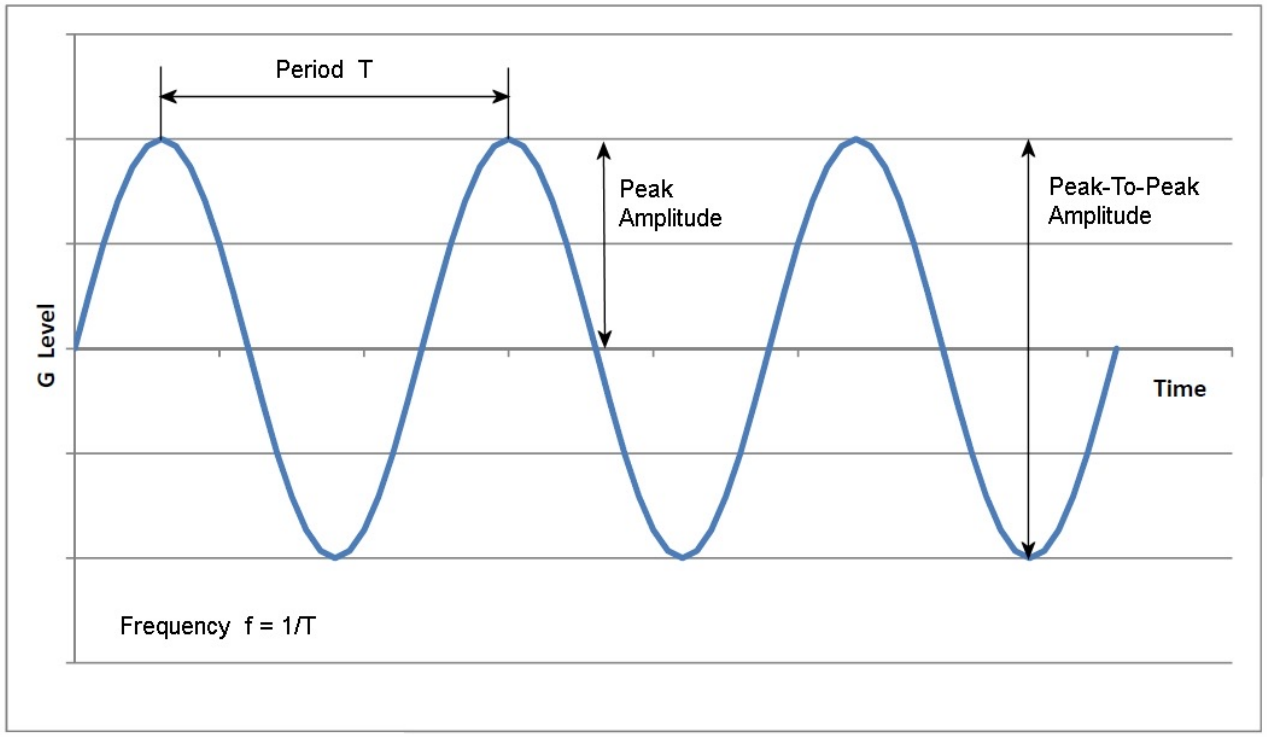
- Amplitude over time
- Oscilloscope





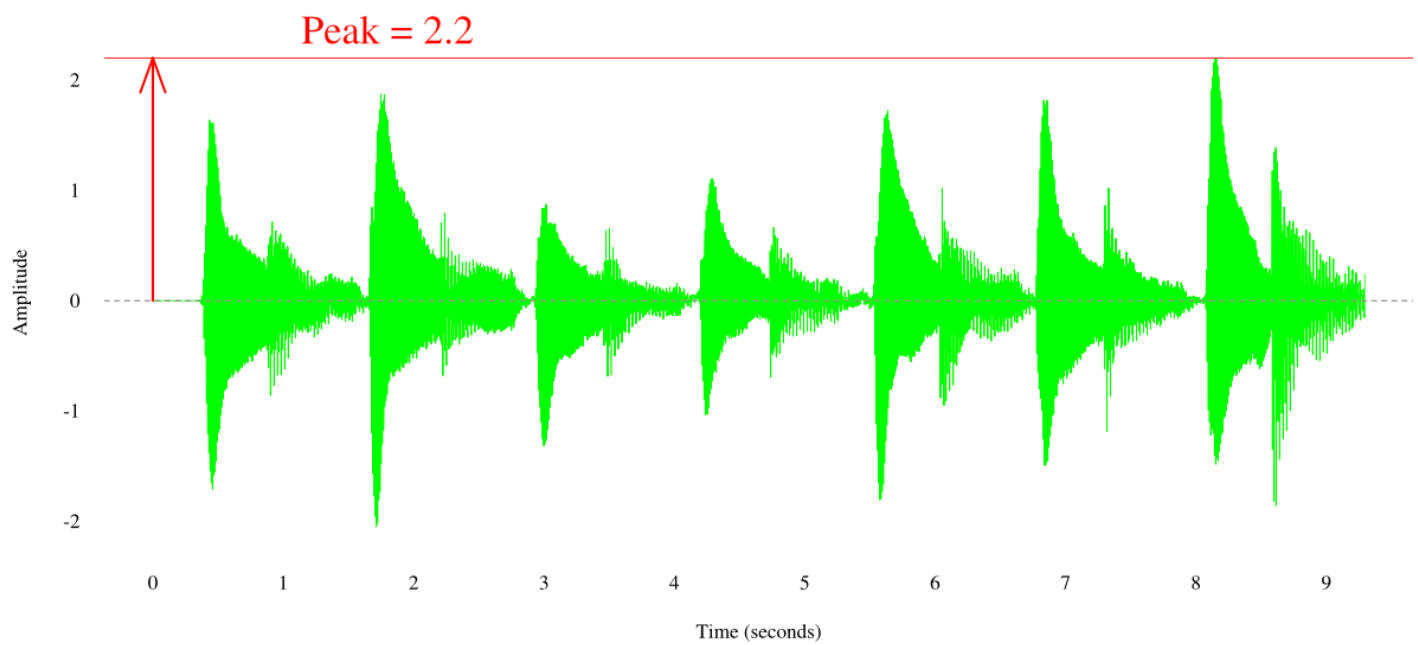
Measuring Amplitude

- Peak Amplitude -> greatest sample over a window*



Measuring Amplitude

- Peak is good for periodic waves, but what about real signals?
- Instantaneous -> exact amplitude quantity at every instant



Measuring Amplitude

- Average across a window
- Root mean square amplitude (RMS)
  - Root of the mean power

$$A_{\mathrm{RMS}}\{x[n]\} = \sqrt{P\{x[n]\}}$$

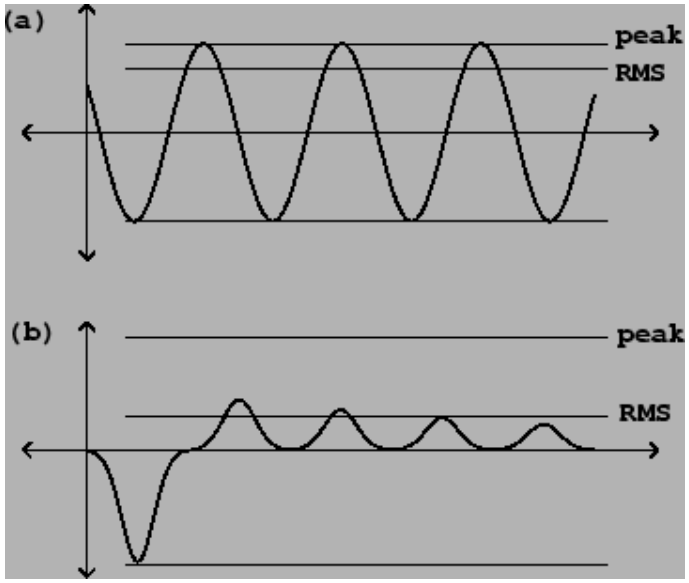
$$P\{x[n]\} = \frac{1}{N}\left(|x[M]|^2 + \cdots + |x[M+N-1]|^2\right)$$

$$RMS = \sqrt{mean(amplitude^2)}$$

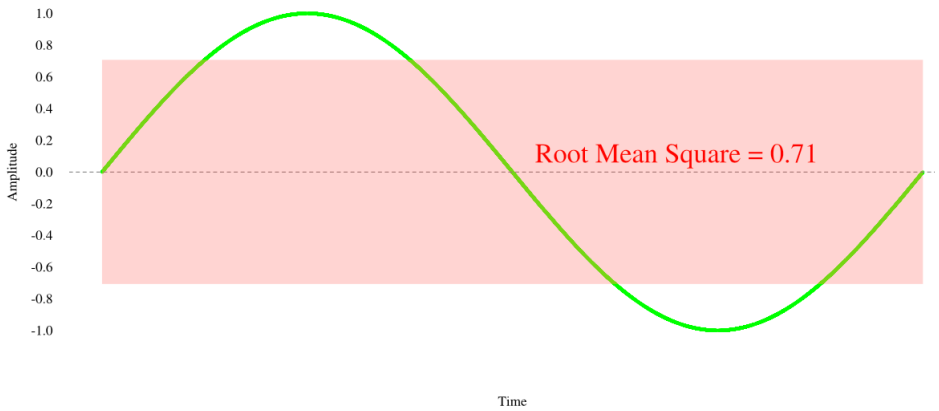
Georgia Tech

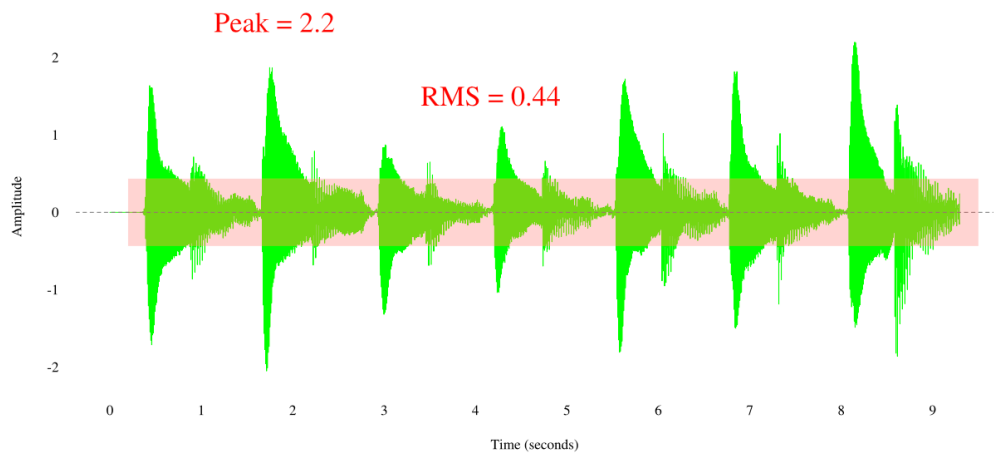
Measuring Amplitude (Cont'd)

- RMS can only be equal or less than peak
- Rule of Thumb for sinusoids : $A_{peak} = a, then\ A_{RMS} = \frac{a}{\sqrt{2}}$
- Crest Factor is the ratio between peak and RMS



Root Mean Square Amplitude

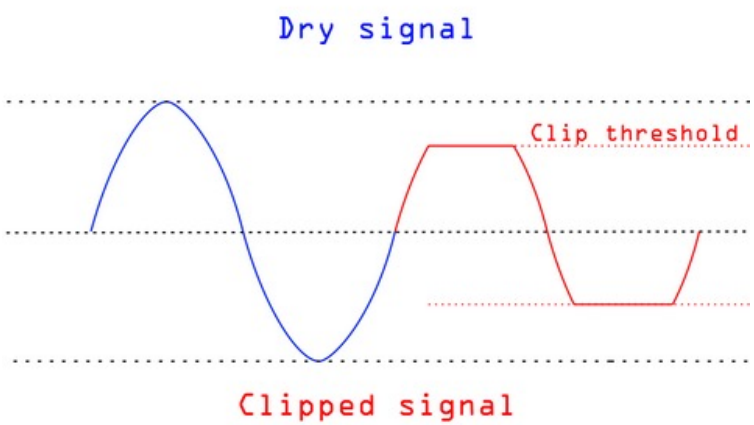




Amplitude Control in Max

Amplitude Range

- Max (and most digital audio systems) expect our maximum amplitude to be 1
- So, a sinusoid will have a dynamic range of -1 to 1
- If we go outside of this range, our system will clip the sound at 1 leading to distortion



Controlling Amplitude

$$x[n] = a\cos(\omega n + \phi)$$

- a can be a constant or a function
  - If we change a slow enough, we can continue thinking of it as a scalar with respect to time
  - Otherwise, we'll use other descriptions…
- Multiply each sample by a factor of y for a new output of $x[n] * y$

Timbre – Amplitude Envelopes

- The change in amplitude over time
- “shape of the sound”
- Usually made up of an Attack, Decay, Sustain, Release

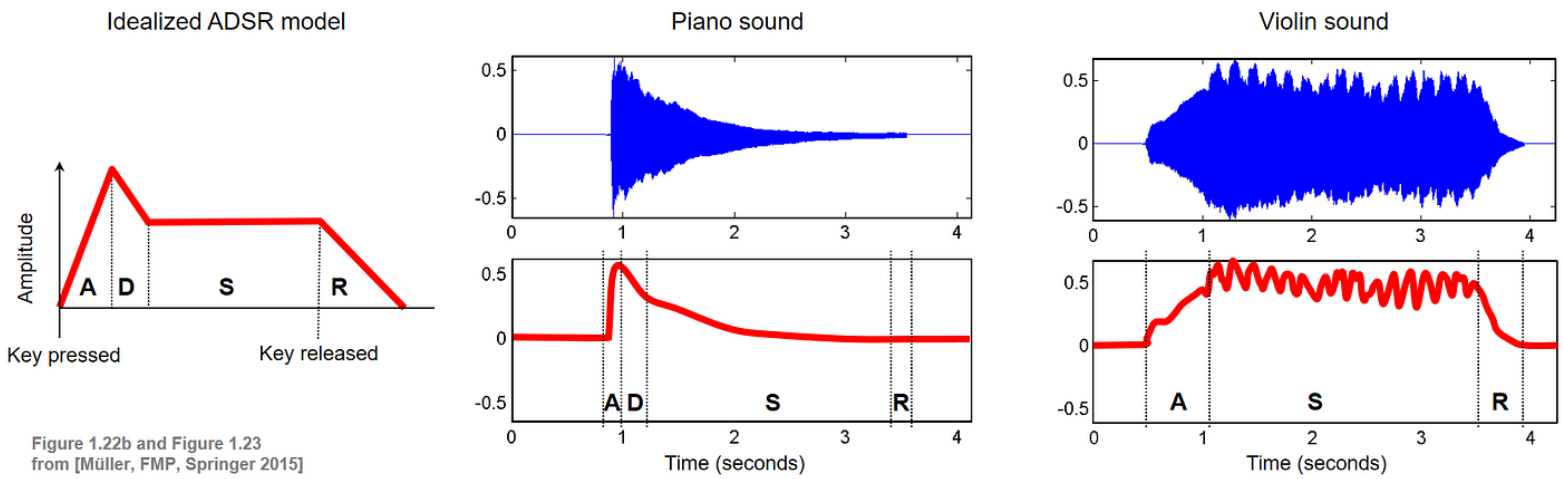
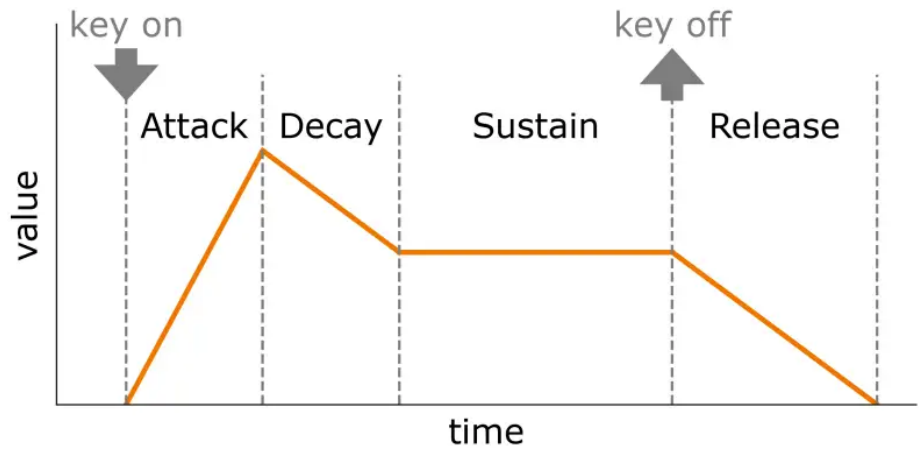


Figure 1.22b and Figure 1.23
from [Müller, FMP, Springer 2015]

ADSR Envelope

- Attack – time it takes to reach maximum level
- Decay – time it takes to reach sustained level
- Sustain – constant level amplitude while a gate is still open (note is held)
- Release – time it takes to decrease to 0



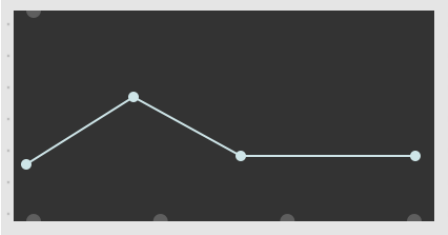
Envelopes in Max

- Line~ creates a linear ramp over time

- Curve~ creates an exponential ramp over time

- Adsr~ creates an adsr signal envelope from a trigger

- Function creates an envelope based on drawn line segments



Up Next…

- Readings/Videos!
- Assignment 1 is due Friday at 11:59pm

Georgia Tech